

# iClusterBayes Summary Report

Automated Report

## Hyperparameters

```
cat('sdev = 0.01, beta.var.scale = 1\n')
```

```
## sdev = 0.01, beta.var.scale = 1
```

## Key Summary Statistics

```
cat(paste0('```\n', '
--- Summary for K = 1 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.6811  0.7061  0.7122  0.7132  0.7206  0.7428

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5739  0.7106  0.6509  0.7961  0.9606
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
    1         1         1         1         1         1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.0000  0.5572  0.6933  0.6350  0.7783  0.9333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
  0.01667 0.04500 0.07167 0.23744 0.26722 1.00000
```

```

Modality 3 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
Modality 4 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02000 0.04556 0.06944 0.20865 0.20583 1.00000
Modality 5 :
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.02111 0.05278 0.13500 0.33937 0.63889 1.00000

--- Summary for K = 2 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.5144  0.5406  0.5489  0.5492  0.5578  0.5861

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3433  0.5267  0.4765  0.6511  0.9333
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.3378  0.5133  0.4593  0.6278  0.8672
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.00000 0.01278 0.35167 0.48834 1.00000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.000000 0.006111 0.031667 0.317261 0.846250 1.000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000  0.5306  1.0000  0.7579  1.0000  1.0000

--- Summary for K = 3 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
 0.3722  0.3967  0.4050  0.4053  0.4133  0.4850

```

```

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2256 0.4094 0.3780 0.5383 0.9178
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.2272 0.4083 0.3694 0.5272 0.8111
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.6483 0.9767 0.7651 1.0000 1.0000
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000000 0.0005556 0.0072222 0.2812454 0.6308333 1.0000000
  Modality 5 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.5844 1.0000 0.7771 1.0000 1.0000

--- Summary for K = 4 ---
Z.ar:
  Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.2750 0.3117 0.3200 0.3206 0.3289 0.3950

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
0.0000 0.1157 0.3156 0.3033 0.4639 0.8711
  Modality 2 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median    Mean 3rd Qu.  Max.
      1      1      1      1      1      1
  Modality 5 :

```

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.1200  | 0.3161 | 0.2964 | 0.4572  | 0.7767 |

| Modality 2 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.4422  | 0.6806 | 0.6310 | 0.9678  | 1.0000 |

| Modality 3 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.9806       | 1.0000  | 1.0000 | 1.0000 | 1.0000  | 1.0000 |

| Modality 4 : |          |          |          |          |          |
|--------------|----------|----------|----------|----------|----------|
| Min.         | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
| 0.000000     | 0.000000 | 0.001111 | 0.264294 | 0.574028 | 1.000000 |

| Modality 5 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.7611  | 0.9817 | 0.8121 | 1.0000  | 1.0000 |

--- Summary for K = 5 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4094 | 0.4467  | 0.4589 | 0.4592 | 0.4722  | 0.5144 |

beta.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.05222 | 0.19167 | 0.21703 | 0.34389 | 0.89222 |

| Modality 2 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 3 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 4 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

| Modality 5 : |         |        |      |         |      |
|--------------|---------|--------|------|---------|------|
| Min.         | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
| 1            | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

| Modality 1 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.00000      | 0.05111 | 0.18556 | 0.20186 | 0.32667 | 0.69778 |

| Modality 2 : |         |        |        |         |        |
|--------------|---------|--------|--------|---------|--------|
| Min.         | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
| 0.0000       | 0.0000  | 0.0000 | 0.2122 | 0.3650  | 1.0000 |

| Modality 3 : |         |         |         |         |         |
|--------------|---------|---------|---------|---------|---------|
| Min.         | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
| 0.06389      | 0.73556 | 0.79833 | 0.78521 | 0.84944 | 1.00000 |

| Modality 4 : |  |  |  |  |  |
|--------------|--|--|--|--|--|
|--------------|--|--|--|--|--|

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.00000 | 0.00000 | 0.09716 | 0.00000 | 1.00000 |

Modality 5 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.2350  | 0.6411 | 0.5681 | 0.9300  | 1.0000 |

--- Summary for K = 6 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.3472 | 0.3861  | 0.3967 | 0.3974 | 0.4072  | 0.4783 |

beta.ar:

Modality 1 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.03222 | 0.20306 | 0.22412 | 0.35722 | 0.86833 |

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

Modality 1 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.03444 | 0.19306 | 0.19845 | 0.32667 | 0.65111 |

Modality 2 :

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.000000 | 0.006667 | 0.247886 | 0.477778 | 1.000000 |

Modality 3 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.9872  | 1.0000 | 0.9591 | 1.0000  | 1.0000 |

Modality 4 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.00000 | 0.00000 | 0.08718 | 0.00000 | 1.00000 |

Modality 5 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.0300  | 0.3039 | 0.3314 | 0.5300  | 1.0000 |

--- Summary for K = 7 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.3572 | 0.3917  | 0.4050 | 0.4050 | 0.4161  | 0.4733 |

beta.ar:

Modality 1 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
|------|---------|--------|------|---------|------|

0.00000 0.01222 0.14222 0.17334 0.28444 0.81111

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

gamma.ar:

Modality 1 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.01333 | 0.14444 | 0.16368 | 0.27778 | 0.62722 |

Modality 2 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.0000  | 0.0000 | 0.1643 | 0.3128  | 1.0000 |

Modality 3 :

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.0000 | 0.7378  | 0.9333 | 0.8576 | 1.0000  | 1.0000 |

Modality 4 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.00000 | 0.00000 | 0.05192 | 0.00000 | 1.00000 |

Modality 5 :

| Min.    | 1st Qu. | Median  | Mean    | 3rd Qu. | Max.    |
|---------|---------|---------|---------|---------|---------|
| 0.00000 | 0.00000 | 0.08056 | 0.22935 | 0.41444 | 1.00000 |

--- Summary for K = 8 ---

Z.ar:

| Min.   | 1st Qu. | Median | Mean   | 3rd Qu. | Max.   |
|--------|---------|--------|--------|---------|--------|
| 0.4694 | 0.5156  | 0.5267 | 0.5259 | 0.5367  | 0.5728 |

beta.ar:

Modality 1 :

| Min.     | 1st Qu.  | Median   | Mean     | 3rd Qu.  | Max.     |
|----------|----------|----------|----------|----------|----------|
| 0.000000 | 0.006667 | 0.105556 | 0.148312 | 0.241528 | 0.821667 |

Modality 2 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 3 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 4 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

Modality 5 :

| Min. | 1st Qu. | Median | Mean | 3rd Qu. | Max. |
|------|---------|--------|------|---------|------|
| 1    | 1       | 1      | 1    | 1       | 1    |

```

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.007778 0.108889 0.136170 0.235000 0.621111
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.06451 0.000000 1.00000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.4206  0.5717  0.6073  0.8156  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.01897 0.000000 1.00000
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.000000 0.000000 0.09502 0.06889 1.00000

--- Summary for K = 9 ---
Z.ar:
  Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.6278  0.7611  0.7789  0.7756  0.7939  0.8361

beta.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008889 0.092778 0.167666 0.243333 0.872222
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
1      1      1      1      1      1

gamma.ar:
  Modality 1 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000000 0.008889 0.084444 0.111240 0.186111 0.513333
  Modality 2 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000000 0.0000000 0.0000000 0.0006049 0.0000000 1.0000000
  Modality 3 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.0000  0.0000  0.0750  0.2115  0.2828  1.0000
  Modality 4 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.
0.000e+00 0.000e+00 0.000e+00 2.126e-05 0.000e+00 3.333e-02
  Modality 5 :
    Min. 1st Qu.  Median      Mean 3rd Qu.      Max.

```

```
0.00000 0.00000 0.00000 0.00157 0.00000 1.00000
', '\n`'))
```

```
## ' '
##
##
## --- Summary for K = 1 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6811 0.7061 0.7122 0.7132 0.7206 0.7428
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5739 0.7106 0.6509 0.7961 0.9606
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5572 0.6933 0.6350 0.7783 0.9333
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.01667 0.04500 0.07167 0.23744 0.26722 1.00000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02000 0.04556 0.06944 0.20865 0.20583 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.02111 0.05278 0.13500 0.33937 0.63889 1.00000
##
## --- Summary for K = 2 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.5144 0.5406 0.5489 0.5492 0.5578 0.5861
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```



```

## 0.0000 0.3433 0.5267 0.4765 0.6511 0.9333
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.3378 0.5133 0.4593 0.6278 0.8672
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.00000 0.01278 0.35167 0.48834 1.00000 1.00000
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.000000 0.006111 0.031667 0.317261 0.846250 1.000000
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.5306 1.0000 0.7579 1.0000 1.0000
##
## --- Summary for K = 3 ---
## Z.ar:
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.3722 0.3967 0.4050 0.4053 0.4133 0.4850
##
## beta.ar:
## Modality 1 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 0.0000 0.2256 0.4094 0.3780 0.5383 0.9178
## Modality 2 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 3 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 4 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
## Modality 5 :
## Min. 1st Qu. Median Mean 3rd Qu. Max.
## 1 1 1 1 1 1
##
## gamma.ar:

```

```

## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.2272 0.4083 0.3694 0.5272 0.8111
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.6483 0.9767 0.7651 1.0000 1.0000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000000 0.0005556 0.0072222 0.2812454 0.6308333 1.0000000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.5844 1.0000 0.7771 1.0000 1.0000
##
## --- Summary for K = 4 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.2750 0.3117 0.3200 0.3206 0.3289 0.3950
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1157 0.3156 0.3033 0.4639 0.8711
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.1200 0.3161 0.2964 0.4572 0.7767
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4422 0.6806 0.6310 0.9678 1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.9806 1.0000 1.0000 1.0000 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.001111 0.264294 0.574028 1.000000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.7611 0.9817 0.8121 1.0000 1.0000

```

```

##
## --- Summary for K = 5 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4094  0.4467  0.4589  0.4592  0.4722  0.5144
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05222 0.19167 0.21703 0.34389 0.89222
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.05111 0.18556 0.20186 0.32667 0.69778
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.0000  0.0000  0.2122  0.3650  1.0000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.06389 0.73556 0.79833 0.78521 0.84944 1.00000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09716 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000  0.2350  0.6411  0.5681  0.9300  1.0000
##
## --- Summary for K = 6 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3472  0.3861  0.3967  0.3974  0.4072  0.4783
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03222 0.20306 0.22412 0.35722 0.86833
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.

```

```

##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.03444 0.19306 0.19845 0.32667 0.65111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.006667 0.247886 0.477778 1.000000
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.9872 1.0000 0.9591 1.0000 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.08718 0.00000 1.00000
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0300 0.3039 0.3314 0.5300 1.0000
##
## --- Summary for K = 7 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.3572 0.3917 0.4050 0.4050 0.4161 0.4733
##
## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01222 0.14222 0.17334 0.28444 0.81111
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.01333 0.14444 0.16368 0.27778 0.62722
##   Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0000 0.1643 0.3128 1.0000

```

```

## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.7378 0.9333 0.8576 1.0000 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.05192 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.08056 0.22935 0.41444 1.00000
##
## --- Summary for K = 8 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.4694 0.5156 0.5267 0.5259 0.5367 0.5728
##
## beta.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.006667 0.105556 0.148312 0.241528 0.821667
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
## Modality 1 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.000000 0.007778 0.108889 0.136170 0.235000 0.621111
## Modality 2 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.06451 0.00000 1.00000
## Modality 3 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.0000 0.4206 0.5717 0.6073 0.8156 1.0000
## Modality 4 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.01897 0.00000 1.00000
## Modality 5 :
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.00000 0.00000 0.00000 0.09502 0.06889 1.00000
##
## --- Summary for K = 9 ---
## Z.ar:
##   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
## 0.6278 0.7611 0.7789 0.7756 0.7939 0.8361
##

```

```

## beta.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.092778 0.167666 0.243333 0.872222
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
##      1      1      1      1      1      1
##
## gamma.ar:
##   Modality 1 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.008889 0.084444 0.111240 0.186111 0.513333
##   Modality 2 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000000 0.0000000 0.0000000 0.0006049 0.0000000 1.0000000
##   Modality 3 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.0000 0.0000 0.0750 0.2115 0.2828 1.0000
##   Modality 4 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000e+00 0.000e+00 0.000e+00 2.126e-05 0.000e+00 3.333e-02
##   Modality 5 :
##   Min. 1st Qu.  Median      Mean 3rd Qu.    Max.
## 0.000000 0.000000 0.000000 0.00157 0.00000 1.00000
##
## ‘‘‘

```

## Conclusion

and basic statistics for each K (1–9). The .png files show Z.ar, beta.ar, and gamma.ar distributions.